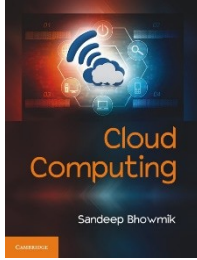


Cloud Computing

Chapter 12

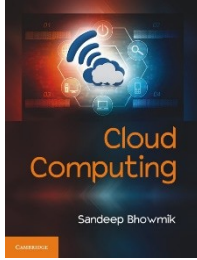
Service Oriented Architecture

Cambridge University Press



Introduction

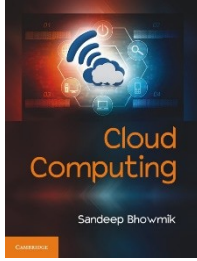
- Management of large system turns out to be a complex task if the architecture of the system is not flexible enough.
- Any large system should be decomposed into functional primitives for efficient and flexible management.
- This philosophy perfectly fits for cloud computing system also.



The Pre-SOA Era

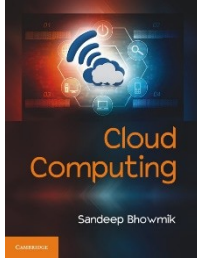
Centrally administrated architecture

- Once, the conventional computing systems were based on centrally administrated architecture.
- But such architecture was not apt for quick application development.
- Neither was it designed to support rapid changes in functionality.
- Moreover, the processes of developing a new application system used to become complex, time-consuming.



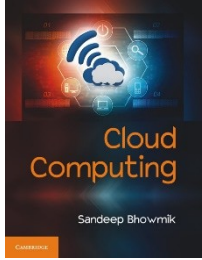
The Pre-SOA Era

- **Component based application development (CBD) model**
- Then emerged the concept of component-based application development (CBD) model.
- CBD promotes decomposition and reusability.
- It also increases productivity, quality, and decreases the time-to-market.



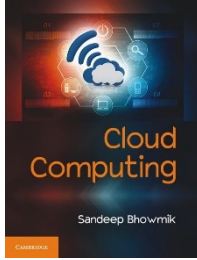
The Pre-SOA Era

- **Component based application development (CBD) model**
- With the widespread adoption of heterogeneous distributed systems, the need of application interoperability appeared.
- *Heterogeneous system* refers to the ability of integrating multiple type (cross architecture and/or cross vendor) of one or more computing resources (like processor) into a single computing environment.
- But interoperability is the issue that could not be handled with CBD model.



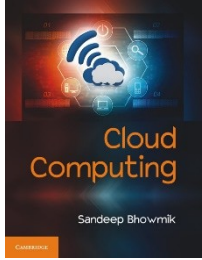
The Cloud Era

- Cloud computing system is the ultimate outcome of heterogeneity.
- Hence an architectural approach was required for applications that could fit in such environment.
- Service oriented architectural approach for application building fulfilled the requirement.



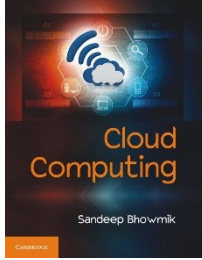
Role of SOA in Cloud Computing

- Cloud computing systems are developed for large scale business delivery and built upon distributed heterogeneous environment.
- Cloud applications are developed over a dynamic infrastructure.
- Service oriented architecture is exactly what was needed to construct the application architecture for such a system.



Role of SOA in Cloud Computing

- SOA approach helps to make applications flexible as well as scalable.
- SOA is a way of building applications.
- Whereas 'cloud' refers to the computing infrastructure as well as the delivery model for that application.

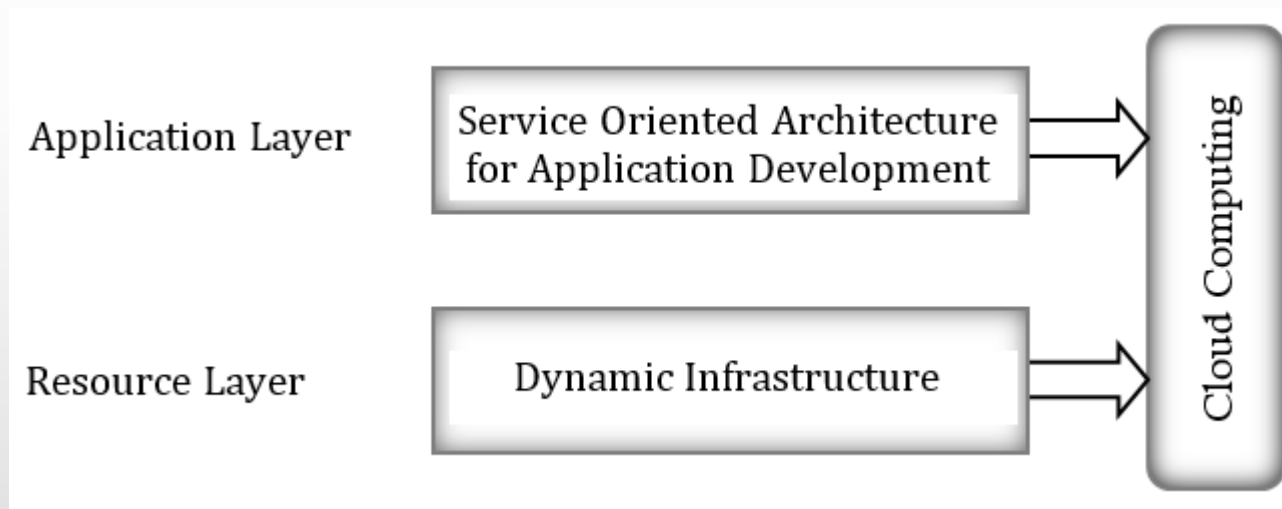


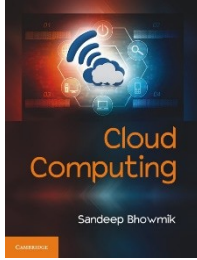
Role of SOA in Cloud Computing

- Many among the objectives of both SOA and cloud are common, such as –
 - Increased flexibility
 - Reduced cost
 - Easier maintenance
- Another crucial role played by SOA is the implementation of loose coupling.

The two layered implementation

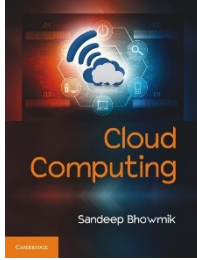
- Together dynamic infrastructure and SOA result in cloud computing





Service Oriented Architecture

- Service Oriented Architecture (SOA) is an approach for application development.
- Software components in SOA paradigm are referred to as services.
- A service is a self-contained functional or business activity having some specific outcome.
- Applications can be built from reusable service components.

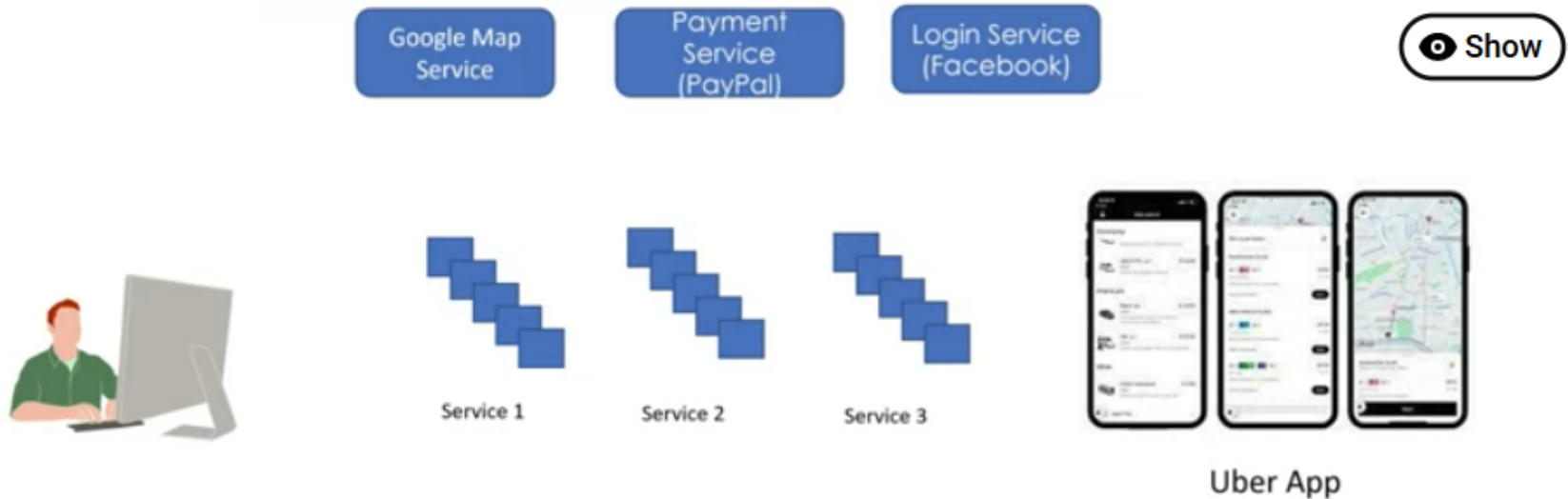


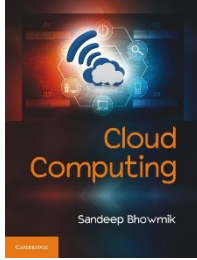
Service Oriented Architecture

- Application development approach based on SOA paradigm allows dynamic construction and rapid customization.
- SOA approach improves agility of application that enables them to respond promptly.
- SOA can be seen as a specification for using and developing service components which are –
 - Language-independent and
 - Platform-independent.

Example

Service Oriented Architecture (SOA)

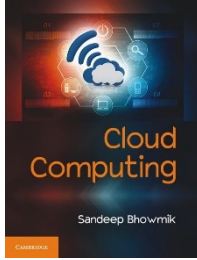




Service Oriented Architecture

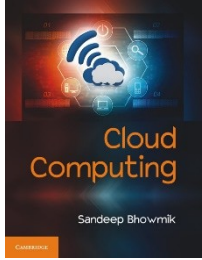
- Few crucial facts

Myth	Fact
SOA is a technology	It is a design philosophy independent of any vendor or technology.
SOA is entirely a new idea	CORBA, DCOM and electronic data interchange (EDI) are examples of earlier service orientation approaches.
SOA implementation needs use of web services	Use of web service is not mandatory for SOA implementation. SOA can be implemented with or without web services.
SOA can be developed	SOA can't be developed; rather it is a mean for application development.



The Emergence of SOA

- In current world, business and IT need to go hand-in-hand.
- The concept of SOA emerged from the necessity to narrow the business-IT gap while building enterprise applications.
- The combination of Business Process Management (BPM) and SOA has closed the gap between business management and IT implementation.

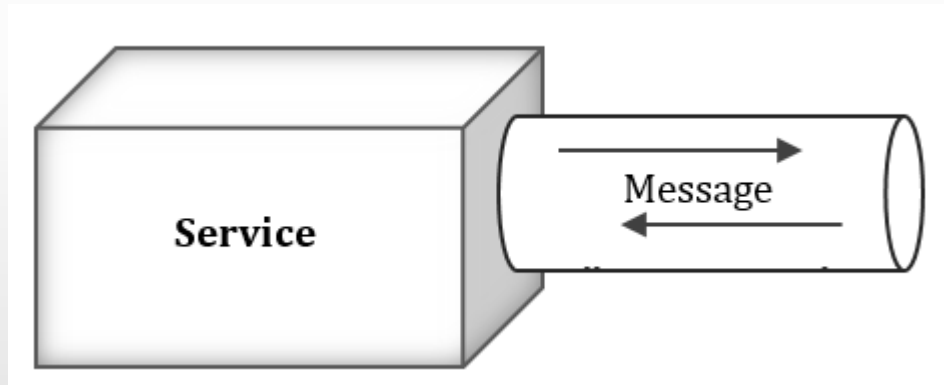


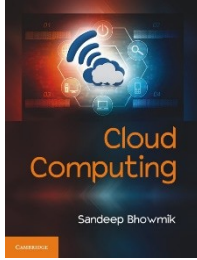
The SOA Architecture

- With SOA approach, applications are built through a collection of services.
- A *service* is an independent entity executing some functionality.
- The services communicate via message passing.

The SOA Architecture

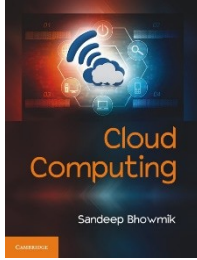
- The model of SOA





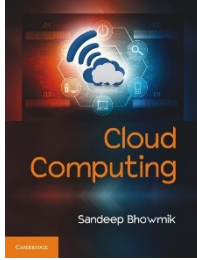
The SOA Architecture

- The message passing schemas of SOA eliminates the technological dependency between communicating agents.
- This allows loose coupling between communicating software agents and promotes interoperability.



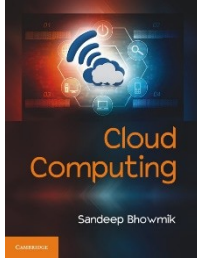
Provider and Consumer

- A service component can play the role of provider or consumer or both.
- Both the roles of provider service and consumer service are played by software agents.
- A consumer service sends a service request to a provider service through message passing.
- The provider service returns a response to the consumer service.



The Interaction Model

- Service providers and service consumers generally do not interact directly with each other.
- A software module is employed to play the role of *broker* who eases the interaction.
- Broker does the act of assembling all published interfaces (of provider services) at one place referred as *service registry*.
- This is also referred as *service directory* or *service catalog*.

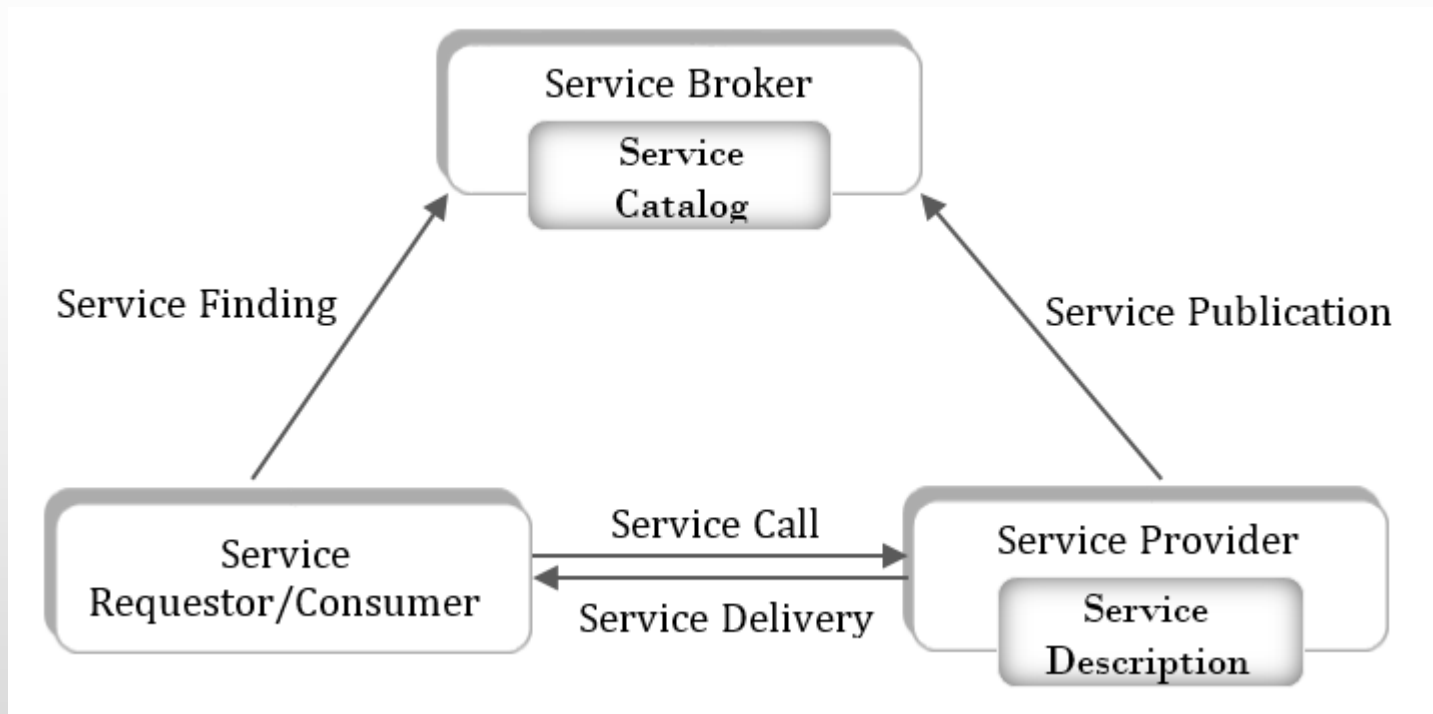


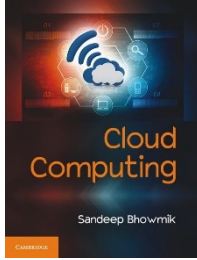
The Interaction Model

- Interaction among services in SOA paradigm works in the following manner:
 - A service provider registers its services in the service registry managed by broker.
 - The service is registered accordingly in registry as per its category.
 - Service consumer looks up for services in the registry.
 - Service consumer binds itself to the service provider and uses the service.

The Interaction Model

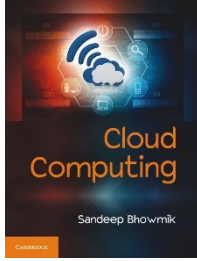
- Service interaction model in SOA





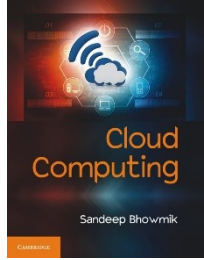
BPM and SOA

- A business is combination of multiple services.
- Alternatively, it can be said that services are part of business.
- The philosophy behind service orientation is to handle the business process management (BPM) more actively.
- SOA is closely attached with BPM where lowest-level activities of a decomposed process are treated as services.



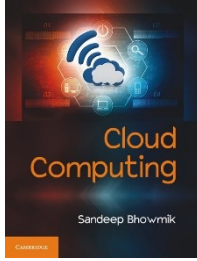
Event Driven SOA

- In older SOA (known as SOA 1.0) approach services were designed as pre-defined processes.
- In event driven SOA (also known as SOA 2.0) events can trigger the execution of activities.
- SOA 2.0 appeared to address all the requirements of dynamic and real-time business.
- An SOA powered with events facilitates agile, adaptive business processes.



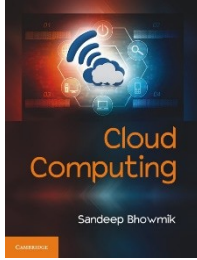
Goal of System Designing

- There are two goals for any superior system designing -
 - De-coupling
 - Composability



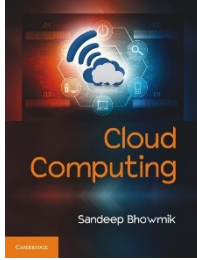
De-coupling

- Coupling between software components should be such that they can be attached effectively, and removed easily as and when required.
- SOA enables this loose coupling feature among application, which allows plugging or unplugging of services effortlessly.
- This de-coupling makes the system flexible, saves time, effort and cost of system management.



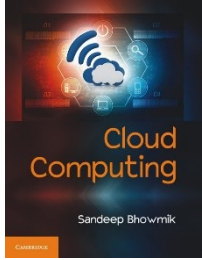
Composability

- A composable component has the following properties –
 - Modular: This means an independent and self-contained component
 - Stateless: During communication each transaction should be separate, independent of any other transactions
- SOA promotes the concept of composable application or system development.



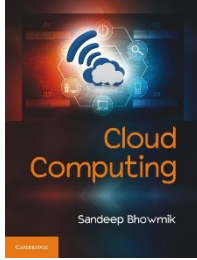
Service Represents Business Functionality

- The idea of a service comes from the world of business, as the world is full of service-oriented businesses.
- A service in SOA paradigm represents discrete business functionality.
- Services in SOA deliver some results that provide value to other services.



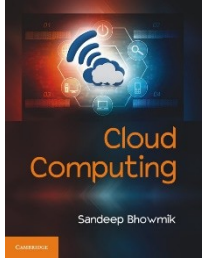
Elements of a Service

- Services have three main components –
 - Contract
 - Interface
 - Implementation



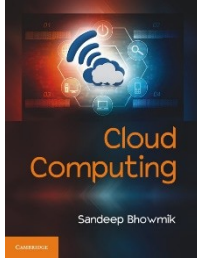
Elements of a Service

- A *contract* specifies the predicted needs of consumer from a service, and details about the service offering of provider.
- The *interface* defines how to access a service and use it.
- Service providers register these interfaces in service registry.
- It is the responsibility of provider to implement a service for optimal performance.



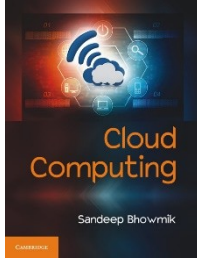
Service Implementation Approaches

- The linking between a service interface and its implementation can be done in three ways –
 - Top-down
 - Bottom-up
 - Meet-in-the-middle



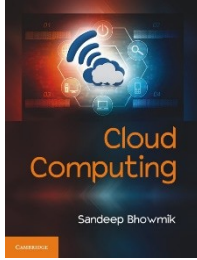
Service Implementation Approaches

- In *top-down approach*, the service interface is defined first, and the implementation follows this definition.
- *Bottom-up approach* works in the opposite manner, where the interface is generated based on an existing service implementation.
- The *meet-in-the-middle approach* takes the best of both top down and bottom up approaches. Here a mapping of the implementation is done to the existing service interface.



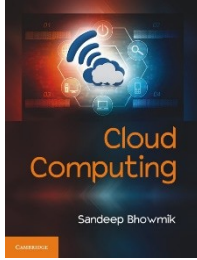
Attributes of Services

- Service has many attributes.
- However, it is not necessary that all these attributes are always required for all the services.
- Depending on requirement, a service oriented developmental approach may implement some of them.



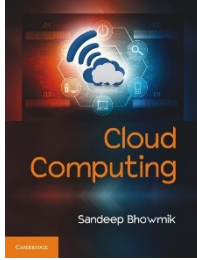
Attributes of Services

- Few among the service attributes –
 - **Self-Contained**
 - One objective of SOA approach is to make a service self-contained or minimize dependency.
 - **Coarse-Grained**
 - Coarse-grained systems consist of fewer number of components which are larger in size.
 - Coarse-grained service can often improve performance in networked environment by reducing switch between service calls.



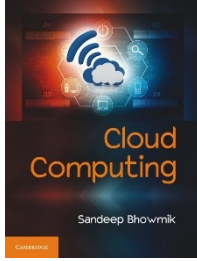
Attributes of Services

- Few among the service attributes –
 - **Stateless**
 - A stateless service is a facility that does not maintain any state between different service calls.
 - **Reusable**
 - Avoiding redundancy is a general goal of software development.
 - Services in SOA are reusable, and just need a call to get effect.



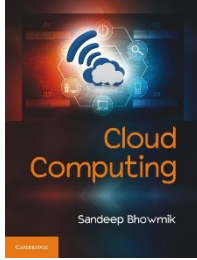
Open Standard Implementation of Services

- As anyone can build service using any technology, it would have been extremely difficult to combine such services without the presence of a proper standard.
- In SOA service development is done following open standards.
- The standards are developed by the Open Group, an international consortium .



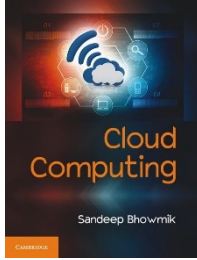
Open Standard Implementation of Services

- **Technologies in Use**
- Extensible Markup Language (XML) - to transport and store data.
- Web Services Description Language (WSDL) – to describe interface.
- Simple Object Access Protocol (SOAP) - a platform and language independent communication protocol.
- Universal Description, Discovery and Integration (UDDI) - as directory service.



Benefits of SOA

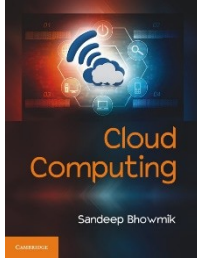
- **Architectural Benefits**
- Flexibility - Changes can be easily implemented.
- Reusability - Services are reusable.
- Simpler integration - reduces complexity of system integration.
- Maintainability - Service maintenance becomes easier.



Benefits of SOA

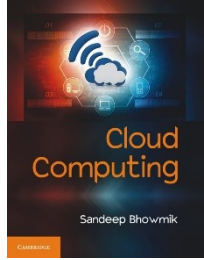
- **Business Benefits**

- **Agility** - Service orientated architecture is based on the assumption that system changes over time.
- **Cost reduction** - Reusability means lower development and testing cost.
- **Increase in quality** - SOA provides high performance and delivers good return on investment (ROI).
- **Quicker release to market** - Service oriented approach takes lesser time for system development.
- **Standardization** - Services in SOA are generally developed following open standards.



SOA and Cloud Computing

- Cloud computing and SOA have many common concerns -
 - Loose coupling
 - Reusability
 - Distribution
 - Flexibility etc.
- Both follow consumer- provider-broker model for interaction.
- Capabilities of cloud are represented as services, and that creates scope for SOA to play important role in cloud computing.
- Thus, they deliver best outcome while working together.



Thank You